Abhishek Kumar

Gurugram, India | abhishek.k.11724@gmail.com | +91-7973373716

linkedin.com/in/abhishek-kumar-ld | github.com/abhishekkumar11724 | abhishekkumar11724.github.io

Summary

Software Development Engineer with 2.5+ years building GenAI agent platforms and distributed backends. Sole developer of the **NL-to-workflow compiler** and backend owner of the **workflow engine** on Delhivery's agent platform, which serves **∼5M workflow and ∼2M agent requests per day**.

Skills

- **Languages**: Python, JavaScript, TypeScript, C/C++; *Familiar:* Go (Golang), GraphQL
- **GenAI & LLMs**: RAG Pipelines, AI Agents, MCP (Model Context Protocol) Servers, LangChain, LangGraph, OpenAI/Gemini APIs, Vector Search
- **Backend & Distributed Systems**: Microservices, Event-driven Architecture, REST API Design, FastAPI, Node.js, Express.js, Kafka, Redis (Bull), AWS SQS, Multithreaded & Async Programming
- **Databases**: MongoDB, PostgreSQL, MySQL, Elasticsearch (Vector), Qdrant, Redis
- **DevOps & Tools**: Docker, Kubernetes, AWS (Lambda, ECS), Git, CI/CD, Linux/Unix, Shell Scripting
- **Practices**: Agile/Scrum, System Design, Design Patterns, OS Internals, Data Structures & Algorithms

Experience

Software Development Engineer, Delhivery – Gurugram, IN — Mar 2025 – Present

- ***NL-to-Workflow Compiler:*** Sole developer of a compile-once engine turning natural-language SOPs into validated workflow graphs with **no LLM in the control loop**, cutting per-query LLM cost **∼90%** and latency to **sub-second**.
- ***Workflow Engine at Scale:*** Built and own the workflow-builder backend: canvas graphs compiled to **LangGraph** (10 node types, parallel branches, pause/resume), serving **∼5M workflow and ∼2M agent requests/day**.
- ***MCP & Agent Platform:*** Developed the platform that turns OpenAPI specs alone into **MCP servers, tools, and agents** (FastAPI, LangGraph, Redis), now **1000+ tools and 150+ agents** across Tech, Finance, and Ops.
- ***Agents & Retrieval:*** Engineered **multi-agent systems** that cut Operations MTTR **∼40%** by automating cross-dashboard retrieval, and a **hybrid search engine** (dense + sparse) with reranking that lifted relevance **∼35%**.
- ***Production Workflows & Tools:*** Shipped non-delivery, shipment-summary, and warehouse workflows (saving **∼Rs. 80 lac**; running cost now **∼Rs. 60K/month**), and OCR, auth-token, memory, and RAG tools, often within a day.
- ***Production Reliability:*** Moved catalog and gateway from a single Uvicorn process to 2-worker **Gunicorn** with request recycling and hung-worker recovery, and fixed a gateway thread lock, cutting **p99 latency ∼68%** (1.8s to 550ms).
- ***Trace Observability:*** Debugged workflow runs whose ∼1,400-span traces were missing from **Langfuse**: 1.38 MB iteration spans were overflowing the SDK's 64 MiB batch limit; cut spans to ∼1 KB so traces render in full.
- ***LLM-as-a-Judge Evaluation:*** Built live evaluations where **LLM judges** score production traces asynchronously after each run (Langfuse evaluators), now running **15+ scores across 100+ agents**.
- ***Agent Memory:*** Benchmarked 6 memory frameworks and built a long-term memory engine (Kafka, Graphiti, Neo4j), now in an A/B release on **10K requests** (80% agents, 20% workflows).

Software Development Engineer, Spyne AI – Gurugram, IN — Jul 2024 – Feb 2025

- ***Event-Driven Webhooks:*** Re-architected a cron-based webhook system into an **event-driven pipeline** (Debezium, Kafka, Redis Bull Queue), achieving **100% SLA** on client deliveries.
- ***Webmine Microservice:*** Led Webmine, a **serverless AWS Lambda** service that auto-scrapes vehicle display pages, improving CSM efficiency **30–40%**, with automated live-visibility checks and multi-level fallbacks.
- ***Partner Integration & Deployment:*** Scaled a CSV-based image-processing system with event-driven architecture and rule-based configs; deployed **Node.js + Kafka microservices on AWS ECS**.

Software Development Engineer Intern, Spyne AI – Gurugram, IN — Jan 2024 – Jun 2024

- ***Performance & Features:*** Cut download processing time **80%** (600s to 120s) and launched an end-to-end wall-replace feature (front-end, back-end, AI) serving **8–9K weekly requests**.
- ***Internal Tooling:*** Built tools that cut ad-hoc client requests **25–30%**, image-classification plugins with real-time reprocessing, and **role-based access control** for the image console.

Education

Dr B R Ambedkar National Institute of Technology, Jalandhar — Dec 2020 – Jun 2024
B.Tech in Information Technology